

Pratik Shirbhate

Lead AI/ML Platform Architect | MLOps & ML Platform Engineering | AWS, SageMaker, Terraform | Enabling Scalable Production ML Systems

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CONTACT

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SUMMARY

I turn fragmented ML experiments into production-grade AI platforms — and I've spent 10+ years doing exactly that across healthcare, travel, and enterprise.

At Jet2, I lead ML Platform Engineering for a 50+ person data science organisation. In the last three years:

- Migrated the entire DS org from Dataiku to a cloud-native MLOps platform on AWS SageMaker — from architecture to go-live
- Cut cloud infrastructure costs by 50% (~£150K/year) through compute rightsizing, lifecycle policies, and VPC optimisation
- Productionised 20+ ML models across Retail Pricing, Marketing Attribution, Fuel Forecasting, and Revenue Risk
- Enabled self-service ML infrastructure for 50+ data scientists and ML engineers, reducing time-to-production significantly
- Built CI/CD, model governance, and observability frameworks from scratch — now the org standard

Before Jet2, I built predictive models in healthcare analytics at Evolent Health and developed my data engineering foundation at Cognizant.

What I'm best at: MLOps & ML Platform Architecture, AWS / SageMaker / Terraform, CI/CD for ML, Model Governance & Monitoring, Cost-Optimised AI Infrastructure, Cross-functional Engineering Leadership

Open to Principal / Staff / Lead roles in ML Platform Engineering, AI Infrastructure Architecture, or Enterprise AI Transformation. Based in Pune; open to remote or hybrid UK/India roles.

PROFESSIONAL EXPERIENCE

Jet2 Travel Technologies Pvt Ltd. (5 years 7 months)

Machine Learning Platform Lead | April 2026 – Present | Pune

Led end-to-end migration of a 50+ member Data Science org from Dataiku to Amazon SageMaker, delivering a production-grade, multi-environment MLOps platform on AWS.

- Designed and architected cloud-native ML platform (SageMaker, Snowflake, Terraform, Azure DevOps) supporting 20+ ML projects across dev, test, and production
- Cut cloud infrastructure costs by 50%+ (~£150K/year) via compute rightsizing, lifecycle policies, and VPC optimization
- Established AWS account structure, IAM governance, and VPC architecture enabling secure self-service for 50+ users
- Standardized CI/CD pipelines, release governance, and Docker/ECR environments — increasing deployment frequency
- Built cost monitoring frameworks giving leadership first-ever visibility into per-project AWS spend
- Led incident management practices (PagerDuty, RCA) improving platform reliability
- Managed and mentored 5–8 ML engineers across UK and India, driving upskilling for 50+ engineers

Technical Lead – Data Science & Machine Learning | April 2023 – April 2026 | Pune

Led MLOps and CI/CD engineering, productionizing 20+ ML models across Retail Dynamic Pricing, Marketing Attribution, Fuel Forecasting, and Revenue Trading Risk.

- Founder of MLE Team; led 6–8 ML engineers, owning model productionization from experimentation handoff to live deployment
- Supported 40+ data scientists across business teams, establishing repeatable deployment patterns
- Built CI/CD pipelines for model training, validation, and deployment — standardizing release processes
- Established MLOps best practices: model monitoring, versioning, and lifecycle management
- Laid foundational ML infrastructure that directly evolved into the org's full-scale ML platform

Senior Machine Learning Engineer | April 2022 – April 2023 | Pune

- Built MLE Team from scratch; deployed scalable machine learning pipelines, transitioning models from research to production
- Developed automated workflows for model training, evaluation, and deployment using cloud-based infrastructure
- Implemented monitoring and logging mechanisms to track model performance in production
- Contributed to early MLOps practices, improving repeatability and maintainability of ML systems

Senior Data Scientist | November 2020 – Apr 2022 | Pune

- Developed and deployed machine learning models solving business problems across the Airline domain
- Performed data analysis, feature engineering, and model evaluation to improve predictive performance
- Collaborated with stakeholders to translate business requirements into data-driven solutions

Evolent Health (2 years 2 months)

Senior Data Scientist | July 2020 – November 2020 | Pune

- Collaborated with US data science team and led India data science team
- Designed, developed, tested, and deployed data science solutions in healthcare analytics

Data Scientist | October 2018 – June 2020 | Pune

- Built and deployed ML models for healthcare fraud detection, patient risk scoring, and claims processing
- Created real-time ML API services using AWS Lambda and serverless architecture

Cognizant (2 years 11 months)

Data Science and Big Data Associate | November 2017 – September 2018 | Pune

Data Science Associate | November 2015 – November 2017 | Coimbatore, Tamil Nadu

- Collaborated with on-site lead data scientist to conduct data analysis and build proof of concepts for data science solutions

MAEER'S MIT College of Engineering

Assistant Professor | July 2015 – November 2015 | Pune

- Taught Engineering Graphics, Fluid Mechanics, and Heat Transfer

TECHNICAL SKILLS

MLOps & ML Platform Architecture, AWS SageMaker, Terraform, LLM Platform Engineering & GenAI Infrastructure, Infrastructure as Code, Snowflake, CI/CD for ML, Model Governance & Monitoring, Docker, Azure DevOps, Python, SQL, Apache Airflow, Apache Spark, GCP (GKE, Compute Engine), Machine Learning (sklearn, H2O AutoML), Deep Learning (TensorFlow), Predictive Modeling, Feature Engineering, Tableau, MongoDB, Git, R

KEY PROJECTS & BUSINESS IMPACT

Jet2 – ML Platform & Infrastructure

- Dataiku to AWS SageMaker Platform Migration: Led end-to-end migration of 20+ DS initiatives to a cloud-native MLOps platform (SageMaker, Snowflake, Terraform, Azure DevOps). Enabled self-service for 50+ users. £150K/year cost savings (50% reduction) via compute rightsizing, lifecycle policies & VPC optimization. Replaced legacy tooling with production-grade CI/CD, model governance, and observability.
- MLE Team & MLOps Practice (Founded): Built the ML Engineering team from scratch (6–8 engineers). Productionized 20+ models across Pricing, Marketing, Fuel & Risk. Established repeatable deployment patterns for 40+ data scientists. Reduced time-to-production significantly.

Jet2 – Travel & Airline Revenue

- Hotel Price Elasticity – Surge Detection: Real-time booking surge detection at hotel occupancy date level. +5M GBP revenue (2024, 20% properties)
- Flight Price Elasticity – One Way Supplement: Real-time dynamic airline pricing model on Dataiku + GKE. +3M GBP revenue (2024)
- Package Only Flights: Ensembled supervised, unsupervised & Pulp optimization models. +30% Holiday Mix for identified flights
- Customer Lifetime Value: KNN, K-Means, and RFM ensemble for targeted marketing (in development)

Evolent Health – Healthcare Analytics

- DRG Recoup (Fraud Detection): Ensembled XGBoost, GMM, IsolationForest, K-Prototype. ~3M USD annual savings (1% fraud reduction)
- Auto Intake (OCR Pipeline): OpenCV, Tesseract, NLP for Prior Authorization fax processing. ~4M USD annual savings (40% manual effort reduction)
- Provider Matching: Real-time ML API (AWS Lambda, TensorFlow) <300ms response. ~100K USD/year savings (5% auto-adjudication improvement)
- Readmission Prediction: Automated ML pipeline (Airflow, sklearn, Keras, MongoDB). 3% fraudulent readmission reduction
- Medicaid Retention Model: Logistic Regression, Random Forest, XGBoost + survival model for churn prediction
- Hospital Visit Classification: ML model for place-of-service prediction to optimize ED utilization
- Patient Risk Score Prediction: Risk scoring model for actuarial pricing & planning

EDUCATION

- M.Tech (Mechanical Engineering – Thermo-Fluids) | IIT Gandhinagar | 2013–2015 | 8.76/10
- B.E. (Mechanical Engineering) | SGBAU, Maharashtra | 2009–2013 | 85% (4th Merit, College Topper)
- HSC | Board of Intermediate Education, Hyderabad | 2007–2009 | 90%
- SSC | Maharashtra State Board | 2007 | 84.76%

CERTIFICATIONS & ACCOMPLISHMENTS

- AWS Certified Machine Learning Engineer – Associate
- Speaker at Analytics Vidhya Datahack Summit 2022
- Google Cloud Professional Machine Learning Engineer
- AI for Medicine Specialization (deeplearning.ai)
- AI for Medical Treatment
- Machine Learning (Udemy)
- Kubernetes (Simplilearn)
- Algorithms & Data Structures (Coursera)
- Programming, Data Structures and Algorithms Using Python
- Divide and Conquer, Sorting and Searching, and Randomized Algorithms
- Participation Certificate at Big Data Analytics Conference